

ASG AIRLINES

End-to-End Data Engineering Pipeline

Notebook Documentation

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M.Sc. Data Science

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1. Notebook Overview and Project Objective

The Jupyter Notebook implements the complete airline data engineering workflow from raw-data ingestion to an analytics-ready CSV dataset. It is designed as a controlled, traceable pipeline: the raw Excel workbook is read, inspected, validated, cleaned, transformed, protected against unnecessary exposure of personally identifiable information (PII), and finally written as a flight-level dataset for Power BI.

Notebook sequence

- Load Python libraries and configure the processing environment.
- Read the four sheets from Airlines.xlsx without modifying the source workbook.
- Inspect structure, data types, missing values, duplicates, and sample records.
- Validate data-quality rules and referential integrity before cleaning.
- Apply only justified cleaning and standardization rules.
- Transform timestamps, duration information, and derived flight attributes.
- Handle PII fields using hashing, masking, or exclusion from the reporting layer.
- Integrate the flight-level information without unsafe joins.
- Create the analytics-ready dataset and calculate the required KPIs.
- Record data-quality results and save flights_analytics_ready.csv.

The notebook therefore acts as both the ETL implementation and the execution record for the project. Each transformation is preceded by validation or a clearly stated business/data-quality decision.

2. Source Data and Initial Setup

The source is Airlines.xlsx, containing four logical datasets: flights, bookings, passengers, and payments. The flights sheet is the primary analytical source. Bookings, passengers, and payments contain transactional or personal information and are validated separately rather than being blindly merged into the final flight-level table.

- flights — flight number, airline, source, destination, departure time, arrival time, and recorded duration.
- bookings — booking, passenger, flight, date, status, seat, passport, and emergency-contact information.
- passengers — passenger identity and demographic/contact fields.
- payments — payment identifier, booking identifier, amount, and payment method.

The notebook creates the processed-data directory when required and keeps the raw workbook read-only. This establishes a clean separation between source data and generated output.

3. Data Loading and Initial Data Inspection

The notebook loads the Excel workbook with Pandas and stores each sheet as a DataFrame. Before transformation, it performs a structural inspection so that problems are visible before any records are changed.

- Checks dataset dimensions and confirms that the expected sheets are present.
- Displays column names and data types for each DataFrame.
- Counts missing values and examines representative sample rows.
- Provides an initial view of the actual values received from the source workbook.

This stage is intentionally descriptive. It does not attempt to repair records. The purpose is to establish what the source actually contains and provide the evidence used by the following validation stage.

4. Data Quality Validation

Validation is performed before cleaning. The notebook measures missing values, placeholder values, identifier quality, duplicates, timestamp consistency, and relationships between tables. The key principle is that a value is not changed merely because it looks unusual; a correction is made only when the rule is supported by the data and the intended use.

Important findings

- 15 exact duplicate rows were found in flights and were identified as true full-row duplicates.

- The flight identifier 6F250 occurs in two different flight records with different route/timestamp information. These records represent reused flight numbers and are therefore retained.
- 41 airline values are missing and 31 contain the literal placeholder UNKNOWN.
- One flight, SJ192, has an arrival timestamp earlier than its departure timestamp, producing an anomalous calculated duration.
- Bookings contain 45 missing status values and 30 INVALID values.
- Payments contain 48 missing amounts and 30 INVALID amounts.
- 36 passenger_id values are reused across records representing different people; the affected records are flagged rather than incorrectly merged.
- Aadhaar values occur in 10-, 11-, and 12-digit forms and require controlled standardization.
- There are 10 missing last-name values in passengers.
- Referential-integrity checks found zero orphaned records for the examined relationships.

5. Data Cleaning and Standardization

The cleaning section applies the decisions established during validation. The notebook keeps an explicit distinction between values that can be standardized with confidence and values that cannot be safely recovered.

Flights

- Exact duplicate rows are removed with `drop_duplicates()`.
- Missing airline values and the literal UNKNOWN placeholder are standardized to Unknown.
- The reused flight number 6F250 is retained because its rows contain different legitimate flight information.
- The anomalous SJ192 timing is retained but marked with `duration_anomaly_flag` rather than being silently corrected.

Bookings and payments

- Missing or INVALID booking statuses are standardized to Unknown while retaining the booking records.
- Missing or INVALID payment amounts are marked with `amount_invalid_flag`; no guessed monetary value is inserted.

Passenger data

- Missing last names are filled with Unknown.
- Aadhaar identifiers are converted to strings and zero-padded to 12 digits where the source lost leading zeros.
- Passenger identifier conflicts are explicitly flagged instead of assuming that the same passenger_id always represents the same individual.

6. Flight Time and Duration Transformation

Departure and arrival fields are converted to proper datetime values. The notebook calculates duration from the timestamp difference and compares it with the recorded duration. This provides a data-quality control for impossible or inconsistent flight timing.

- Flight date is derived from the departure timestamp.
- Calculated duration is used to identify timing anomalies.
- An overnight-flight indicator is derived when arrival occurs on a later calendar date.
- The anomaly flag allows problematic records to remain traceable without contaminating KPI calculations.

This approach preserves the source record while separating trustworthy analytical measures from known data-quality exceptions.

7. PII Handling and Data Integration

The notebook identifies personal information in passenger and booking data and prevents unnecessary exposure of these fields in the analytics-ready flight dataset. SHA-256 hashing is used for selected identifiers where a non-reversible representation is useful; other identifying fields are excluded from the final reporting layer.

PII controls implemented in the notebook

- Aadhaar and passport identifiers are protected through one-way hashing where required.
- Passenger names, contact details, and other direct identifiers are not carried into the final flight-level analytics table.
- PII handling occurs before the reporting output is produced.

The notebook also evaluates relationships between the source tables. Although bookings reference flights and passengers, the data contains conditions that make a naïve merge unsafe. In particular, `flight_id` is reused, so joining bookings to flights by `flight_id` alone could duplicate booking counts across multiple flight records. Passenger identifiers also contain conflicts for some records.

8. Analytics-Ready Dataset Creation

The final analytical table is deliberately built from the cleaned flights DataFrame. A route field is created by combining source and destination. Additional analytical fields include flight date, transformed duration information, overnight-flight status, and duration anomaly status.

- One row represents one flight record after exact duplicate removal.
- The table is suitable for aggregation by airline, route, date, and anomaly status.
- Known timing anomalies remain visible through a dedicated flag.
- Transactional and passenger-level data are not forced into the flight-level table.

This design produces a stable grain for Power BI and avoids misleading metrics caused by unreliable joins or repeated identifiers.

9. KPI Preparation

The notebook prepares four required analytical measures directly from the analytics-ready flight dataset. Records flagged for duration anomalies are excluded from the average-duration calculation so that an impossible timestamp relationship does not distort the KPI.

- Average Flight Duration — calculated from valid flight-duration records.
- Route-wise Traffic — counts flights by the derived source-destination route.
- Delays / Anomalies — summarizes the records affected by data-quality timing anomalies.
- Distribution of Flights by Airline — counts flight records for each standardized airline value.

10. Data Quality Summary and Output

The notebook maintains a running data-quality summary so that the final results are traceable to the measurements performed earlier. This includes duplicate counts, identifier issues, relationship checks, and the number of records affected by the cleaning rules.

Final quality decisions

- Only exact duplicate flight rows are removed.
- Legitimate reused flight numbers are preserved.
- Missing and placeholder categorical values are standardized where a reliable replacement exists.
- Unrecoverable numeric values are flagged instead of being fabricated.
- Identifier conflicts are recorded instead of forcing an incorrect identity match.
- Referential integrity remains intact for the checked relationships.
- PII is excluded or protected before the reporting dataset is produced.

11. Saving the Processed Dataset

The final DataFrame is written to `data/processed/flights_analytics_ready.csv`. The notebook verifies the output after saving, providing a final execution check that the expected analytics-ready file has been created.

The generated CSV becomes the hand-off point between the data engineering pipeline and Power BI. The raw Excel workbook remains unchanged, while the processed file contains the cleaned and derived fields required for analysis.

12. Overall Notebook Flow

ASG Airlines — End-to-End Data Engineering Architecture



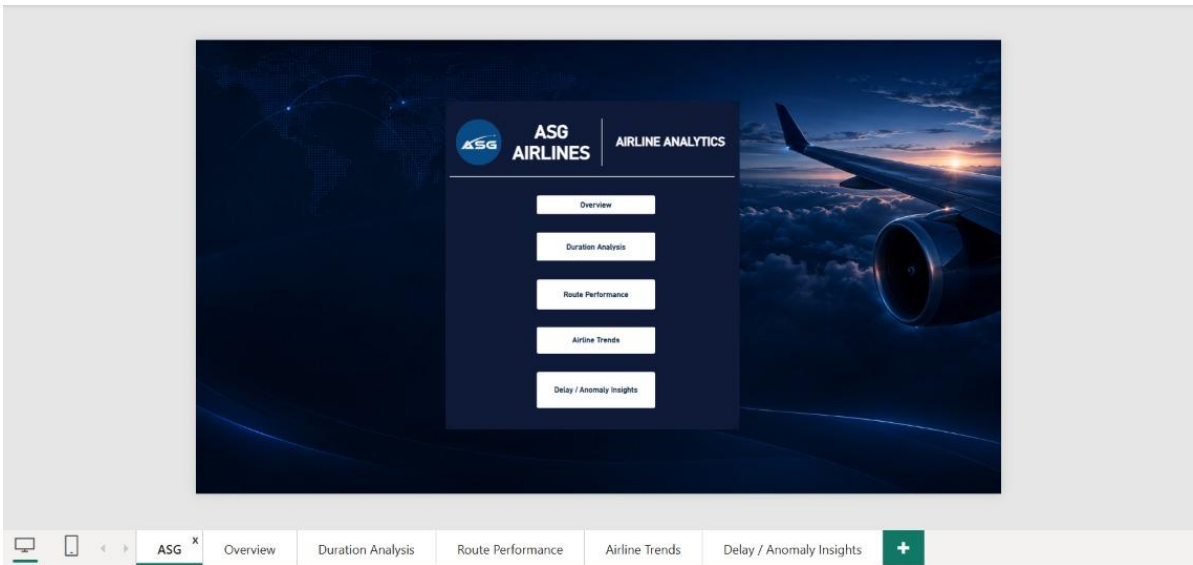
13. Conclusion

The notebook provides a complete, reproducible data engineering workflow for the ASG Airlines use case. Its main strength is controlled transformation: data is first measured, then cleaned according to explicit rules, and finally shaped into a reporting-friendly flight-level dataset. The resulting CSV can be consumed by Power BI for dashboarding while retaining visibility into important data-quality exceptions.

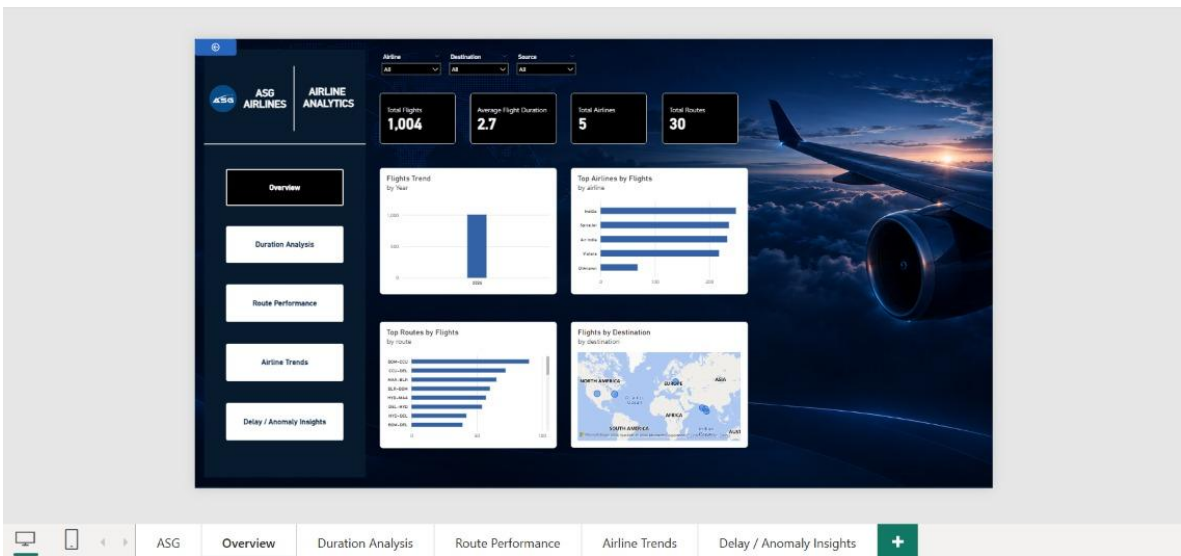
The implementation demonstrates practical data engineering principles including source preservation, profiling, validation-first cleaning, schema and identifier checks, anomaly flagging, PII protection, safe data integration, KPI preparation, and traceable output generation.

Power BI Screenshots

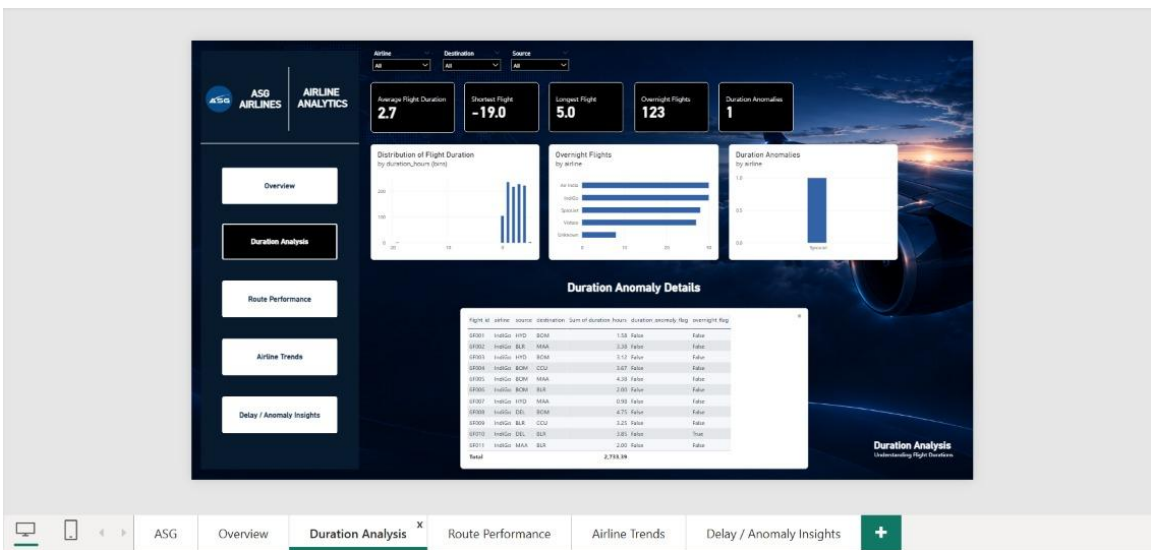
1.Main



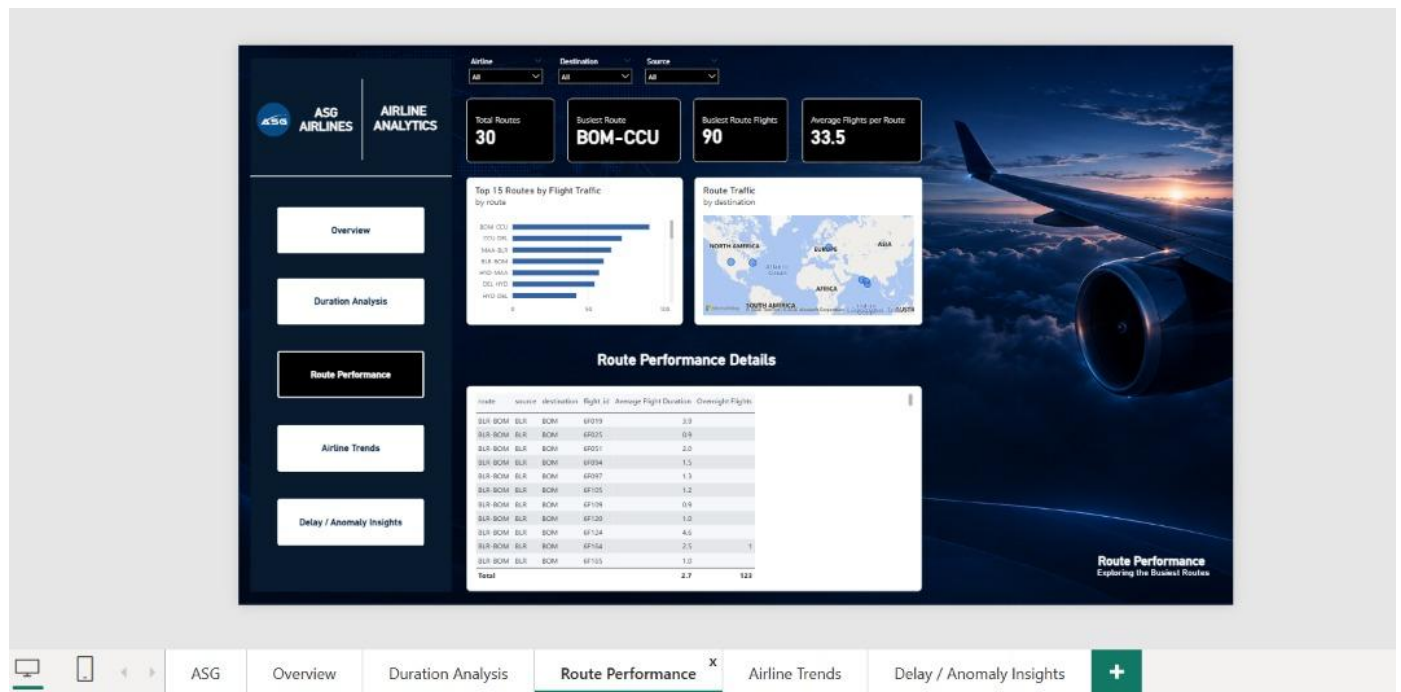
2. Overview



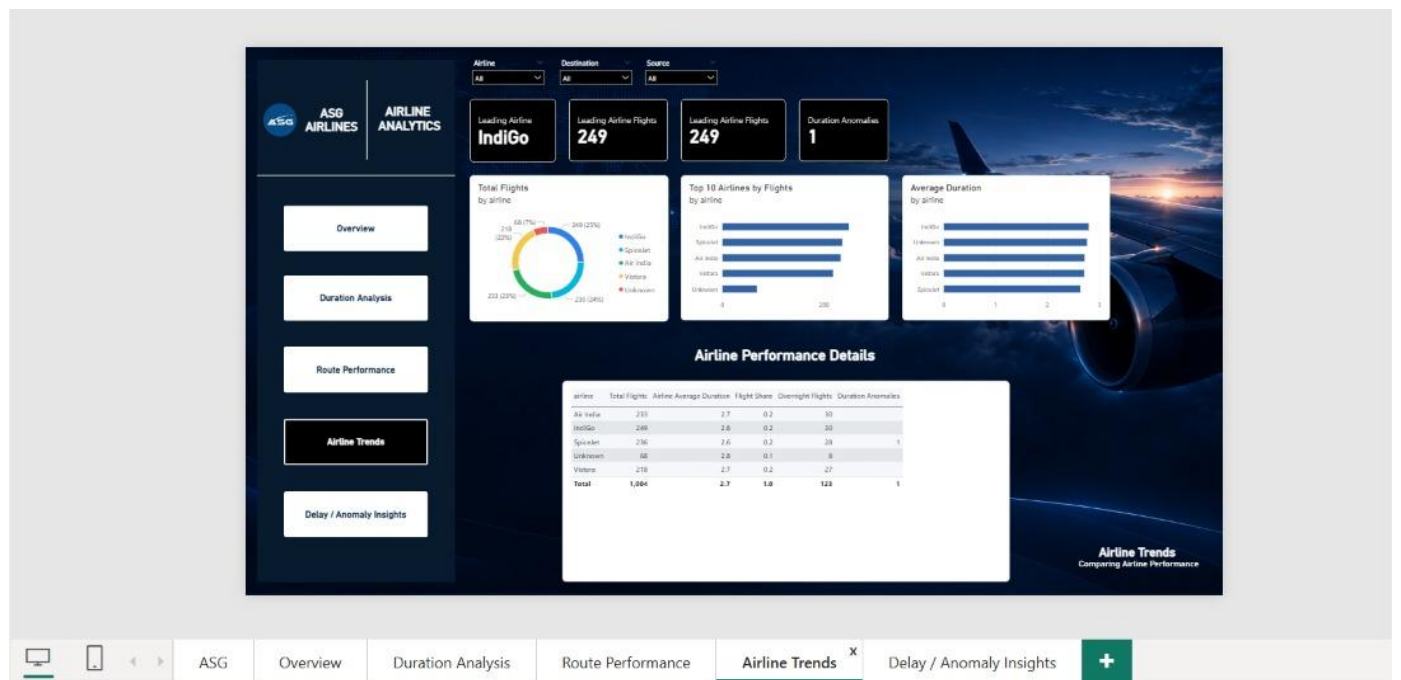
3. Duration Analysis



4.Route Performance



5.Airlines trends



6.Delay/anamoly insights

